
DOMAIN ADAPTATION TECHNIQUES FOR CAR DETECTION USING SYNTHETIC DATA

A PREPRINT

December 4, 2025

ABSTRACT

This project investigates three different domain adaptation techniques to improve objection detection performance when models are trained on synthetic images and evaluated on real-world images. Synthetic data is appealing due to its low cost and perfect annotations, but models trained on such data typically generalize poorly to real world images due to domain shift. We implemented and compared three adaptation strategies: A lightweight CycleGAN to convert synthetic images into realistic versions before training; a Domain-Adversarial Neural Network (DANN) branch integrated into YOLO, enabling feature-level domain alignment; a Teacher-Student (self-training) framework that leverages pseudo-labels from a teacher detector to progressively adapt the student model to real images. Using a synthetic car-detection dataset as the source domain and a real-world driving dataset as the target domain, we focused on improving the performance of a baseline YOLO object detection model trained exclusively on synthetic images.

Keywords Deep Learning · Convolutional Neural Networks · Domain Adaptation · YOLO · CycleGAN

1 Introduction

We explored three fundamentally different approaches to tackle this synthetic-to-real domain gap. Each method targets a different part of the learning pipeline, allowing us to compare multiple strategies rather than relying on a single adaptation mechanism. For image-level adaptation, we used a CycleGAN-style generator paired with a PatchGAN discriminator [1]. The generator is based on a ResNet-style architecture with downsampling, residual blocks, upsampling layers and skip connections [4][2]. Other approach is to adapt the pre-trained YOLO architecture [17] to be compatible with a DANN-based domain adaptation model [18]. This approach allows the resulting network to learn features that would be applicable to real data by introducing a subset of unlabeled real data into the training process. This mechanism, implemented through a special layer, allows the network to learn domain-invariant features during backpropagation. For label-free adaptation of the detector on the target domain, we implemented a Teacher-Student self-training framework. A teacher YOLO model, trained only on synthetic images, is first used to generate pseudo-labels on unlabeled real-world images. Therefore, we train a student detector on a mixture of real images with these pseudo-labels and the original synthetic data, while carefully filtering out low-confidence and very small predictions. This approach allows the detector to progressively adapt to real-world appearance statistics without any manual annotation on the target domain, following the general idea of teacher-student domain adaptation frameworks used in recent work on cross-domain object detection [13].

2 Methods

2.1 The Datasets

The DITM dataset consists of synthetic images generated from the SIM10K dataset, designed specifically for car detection tasks [5]. The dataset includes 10,000 images with bounding box annotations for cars. These images are rendered in a simulated environment, featuring consistent lighting, uniform textures, and minimal background

complexity. The dataset is structured in the VOC2012 format, with annotations stored in XML files and corresponding images in JPEG format. The bounding box annotations focus exclusively on cars, making this dataset ideal for training object detection models in a controlled synthetic environment. Furthermore, the KITTI dataset is a real-world dataset collected from driving scenes, featuring diverse environments, lighting conditions, and weather variations [6]. It includes 7,481 training images and 7,518 testing images, with high-resolution captured from a moving vehicle. This dataset also provides bounding box annotations for cars, stored in plain text files. The images are stored in PNG format and are divided into training and testing folders. The KITTI dataset serves as the target domain for evaluating the domain gap and testing the performance of models trained on synthetic data. Accordingly, for this project, the dataset was split into 70% training, 15% validation, and 15% testing subsets.



Figure 1: 4 Sample images from the DITM Dataset



Figure 2: 4 Sample images from the KITTI Dataset

2.1.1 Difficulties faced with KITTI and DITM

To enable faster training and testing, we reduced the DITM dataset by a factor of 10, selecting 1,000 images for training, validation, and testing [5]. This reduction was made in order to accommodate computational constraints, thereby enabling the training process to be completed within a reasonable time frame. Concerning the KITTI dataset, due to the absence of ground truth labels in the validation and test sets, some adjustments were mandatory [6]. As the test set does not provide public labels, the training set was partitioned into smaller subsets for validation and testing, enabling performance evaluation despite the limited data. Although these modifications were necessary, it is acknowledged that they may introduce certain limitations, as the reduced dataset size and re-partitioning could potentially affect the final results.

3 Lightweight CycleGAN Implementation

3.1 Choice for a CycleGAN

Due to a lack of computational power, we decided to implement a viable, lightweight model that could, in fact, alter images, without changing the image in relatively big proportions. As a matter of fact, we feel the need to justify the choice to implement a CycleGAN, instead of a Variational AutoEncoder or even a Diffusion Model. As mentioned before, the high computational costs and the training complexity drifted us away from the Diffusion Models, as they require careful tuning of noise schedules and mainly excel in creating entirely new images, which is not our goal [8][9]. Furthermore, Variational AutoEncoders were also disregarded due to the probabilistic nature of their reconstruction loss, that often generates blurry outputs [7].

3.2 Implementation

The primary objective of this implementation is to develop a model capable of introducing different types of noise and distortions, simulating conditions found in real-world scenarios [4][1]. To achieve this, we adopted a ResNet-inspired architecture that comprises an initial convolutional block, followed by a downsampling stage, a sequence of residual

blocks, an upsampling stage, and a final output layer [4]. This structure is particularly advantageous for domain adaptation, as it allows for effective image transformation while preserving essential content. Indeed, the network is initialized with a 7x7 convolution, succeeded by Instance Normalization and a ReLU activation function. Instance Normalization was preferred over Batch Normalization to maintain the unique characteristics of individual images, an important factor for style transfer and domain adaptation tasks [4]. Moreover, downsampling is realized through two stride-2 convolutional layers, which reduce feature map dimensionality and contribute to efficient hierarchical feature extraction. At the core of the model lie ten residual blocks. Each block consists of 3x3 convolutional layers, combined with Instance Normalization and ReLU activations. Incorporating skip connections, fundamental to ResNet architectures, into these blocks preserves image details and facilitates more effective gradient flow during training. Therefore, this design enables the generator to learn complex transformations without introducing excessive computational overhead [4]. The upsampling stage is implemented through two transposed convolutional layers, which are tasked with restoring the original spatial dimensions of the images. Again, after each upsampling layer, Instance Normalization and ReLU are applied to ensure stability throughout the training process. The final output layer employs a 7x7 convolution with a Tanh activation function, producing normalized outputs in the range $[-1, 1]$. To sum up, the generator is designed to balance model capacity and efficiency, leveraging its compact structure, moderate number of residual blocks, and consistent use of Instance Normalization to ensure robust performance for domain adaptation [4, 11].

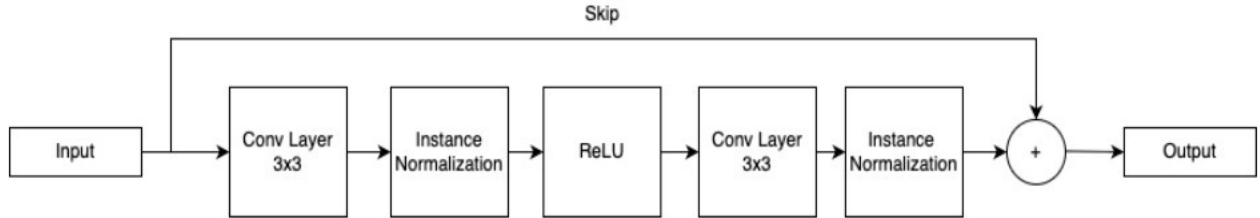


Figure 3: Skip Connections Implementation in the Lightweight CycleGAN

3.3 Results and Final Considerations

Our experimental findings indicate that the GAN-generated dataset does not universally degrade performance, but instead alters it in different ways. Precision increased substantially from 0.2875 to 0.42, showing that the GAN-transformed data led the model to produce more accurate positive detections. However, recall decreased from 0.0911 to 0.0763, meaning the model detected slightly fewer total objects. Similarly, mAP50 rose from 0.0595 to 0.0856, and mAP50-95 increased from 0.0259 to 0.044, indicating modest improvements in overall detection quality. Despite these improvements, the results highlight challenges inherent to the domain-adaptation process applied through the GAN. A primary contributing factor is the instability of GAN training: the generator and discriminator constantly compete, the generator may fail to produce transformations that reliably preserve structural details, especially for small objects such as cars. This can result in distorted or inconsistent outputs. In addition, the lightweight architecture of the generator limits its ability to capture fine-grained transformations, leading to the loss of delicate object features during style adaptation. In several cases, the GAN introduced noticeable distortions in small objects, making cars appear blurred or blob-like. Since accurate object detection depends on precise local detail, this degradation in structural fidelity can lower the model’s ability to detect objects consistently. Finally, the small training dataset restricts the GAN’s ability to learn stable cross-domain mappings: GANs typically require large, diverse datasets for robust performance, and the reduced sample size amplified training instability and limited generalization [10, 11].

4 Domain Adversarial YOLO Implementation

4.1 Related work

Research on domain adaptation had a significant increase in recent years. One of the mentioned reasons is that, in certain areas, it is much easier to create or get large amounts of synthetic data rather than real world data. A work in domain adaptation, Domain-Adversarial Training of Neural Networks [18], introduces a new architecture for learning domain invariant and discriminative features by introducing a gradient reversal layer, to produce embeddings that are indistinguishable between source and target domains. Several other works followed up on DANN principles, such as A Robust Learning Approach to Domain Adaptive Object Detection [15], which shows that DANN methods significantly improve vehicle detection, and UDANet: An unsupervised domain adaptive vehicle density estimation network based on joint adversarial learning [19] which adds two domain adaptive components to a pre-trained backbone, and Domain

Adaptive YOLO for One-Stage Cross-Domain Detection by [16] which wanted to achieve real time performance on an idea proposed in Domain Adaptive Faster R-CNN for Object Detection in the Wild [20], where added domain adaptation components for instance level and image level classification used gradient reversal layer and consistency regularizer was incorporated into these two classifiers to learn domain invariant features.

4.1.1 DANN

Likewise, the core idea behind implementing DANN is to learn domain-invariant features through adversarial training. The architecture consists of three main components, which are feature extractor, label predictor, and domain classifier. First, the feature extractor transforms the input images into latent representations for both domains. The label predictor is trained on labeled data from the source domain for object detection. Finally, the domain classifier receives the same feature embeddings but is trained to differentiate between the source and target domain, however, the gradient of the classifier is reversed before flowing into the feature extractor [18].

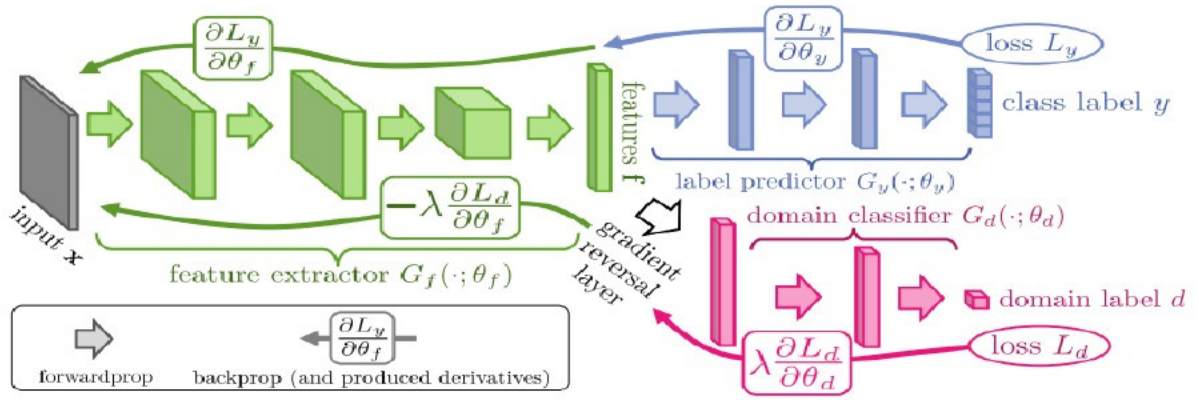


Figure 4: DANN implementation diagram [18]

The reversed gradient forces the feature extractor to produce embeddings which confuse the domain classifier and help the label predictor minimize detection loss. In summary, the result of this is that the feature space becomes domain-invariant, which improves generalization to the target domain [18].

4.2 Implementation

The fundamental concept of the implementation is to integrate supervised YOLO [17] detection training on a labeled source domain with adversarial domain adaptation using unlabeled target data. During the training phase, each batch comprises annotated source images and unannotated target images, both of which are resized and processed identically. Henceforth, a forward hook is attached to an intermediate YOLO backbone layer to extract dense feature representations for both domains without modifying YOLO’s internal code. These features are processed by a Gradient Reversal Layer (GRL) [18] and a lightweight domain discriminator network. This layer is responsible for ensuring that the gradients flowing back into the YOLO backbone are reversed, thereby encouraging the feature extractor to make the source and target features indistinguishable. The computation of supervised YOLO loss is exclusive to source-domain predictions, with target-domain images contributing exclusively to the adversarial domain-alignment loss. The ultimate objective is a weighted sum of detection loss and domain-classification loss, enabling the network to maintain detection accuracy while simultaneously learning domain-invariant features. However, due to the lack of documentation of the internals of YOLO, changing the training process by directly attaching the data to the internal structure proved to be a greater challenge than anticipated. This resulted in dropping the idea due to time constraints and lack of clear solutions.

5 Teacher-Student Self-Training for Domain Adaptation

5.1 Teacher-Student Framework Overview

To further bridge the synthetic-to-real domain gap beyond image-to-image translation and feature alignment, we incorporated a teacher-student self-training framework. In this approach, a detector trained on the synthetic domain

acts as the teacher, generating pseudo-labels on real-domain (unlabeled) images; these pseudo-labels are then used to train a student model. This strategy allows leveraging large quantities of real data without manual annotation, and has recently shown promising results in domain adaptation for object detection. Our process is inspired by the general semi-supervised object detection framework introduced in Unbiased Teacher for Semi-Supervised Object Detection (Liu et al., 2021) which demonstrated that a slowly progressing teacher updated via exponential moving average (EMA), along with careful pseudo-label selection, significantly improves detection performance even with limited labeled data. [12] For domain adaptation specifically, Cross-Domain Adaptive Teacher for Object Detection (Xu et al. 2022) extended this idea by combining pseudo-labeling with adversarial feature alignment and strong/weak augmentations in order to reduce domain shift. [13] We followed that paradigm but took a simplified version to suit synthetic-to-real adaptation under limited computational resources.

5.2 Pseudo-Label Generation on Target Domain

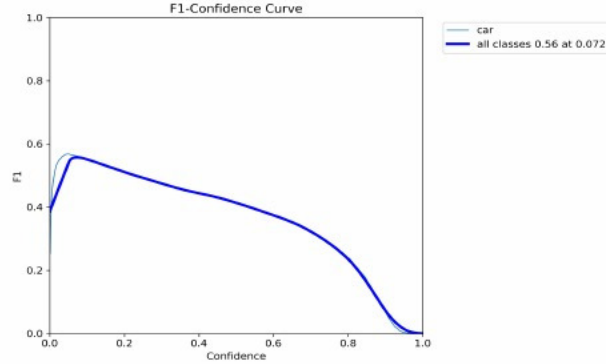


Figure 5: F1–Confidence curve used for selecting the pseudo-label threshold in the teacher–student framework.

In our implementation, the teacher model was trained solely on synthetic data, first processes the target (real) images and outputs predicted bounding boxes with confidence scores. To ensure high-quality pseudo-labels, we empirically determined a confidence threshold of 0.07, corresponding to the peak in the F1-confidence curve (F1 0.56). At this threshold, the balance between precision as well as recall was optimal: many correct detections were retained while suppressing false positives (Figure 5). The selected high-confidence detections were converted into YOLO-style annotation files and merged with the original synthetic dataset, forming an augmented training set combining labeled source data and pseudo-labeled target data.

5.3 Student Training Strategy

During student training, we mixed the synthetic images (with real labels) and the pseudo-labeled real images. The student model was trained for 10 epochs, balancing practicality with computational resources. Over the course of training, we observed a consistent decrease in all loss components (box regression, classification, and distribution focal loss), and improvement in detection metrics despite the domain gap (see Figure 6). The final adapted model achieved $mAP@0.5 = 0.528$, with precision around 0.73 and recall around 0.45, indicating a clear gain over the synthetic-only baseline. The resulting Precision–Recall curve is shown in Figure 7. Confusion matrix analysis (Figure 8) showed that, even though most background patches remained correctly identified, a big part of the real cars remained undetected, highlighting that recall remains a challenge in cluttered or heavily occluded scenes.

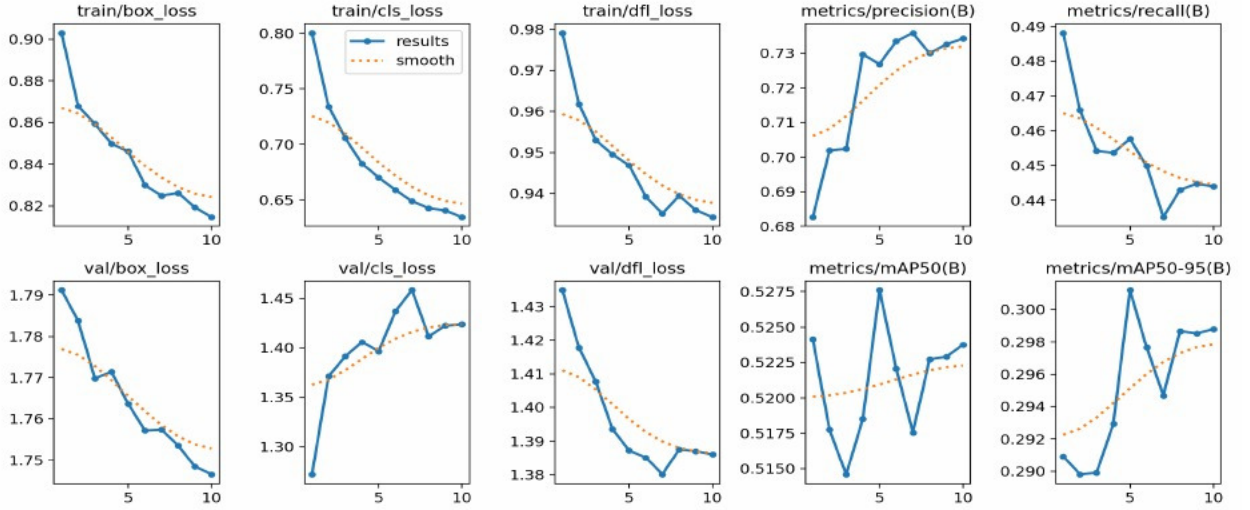


Figure 6: Training and validation losses, precision, recall and mAP curves during Student model training.

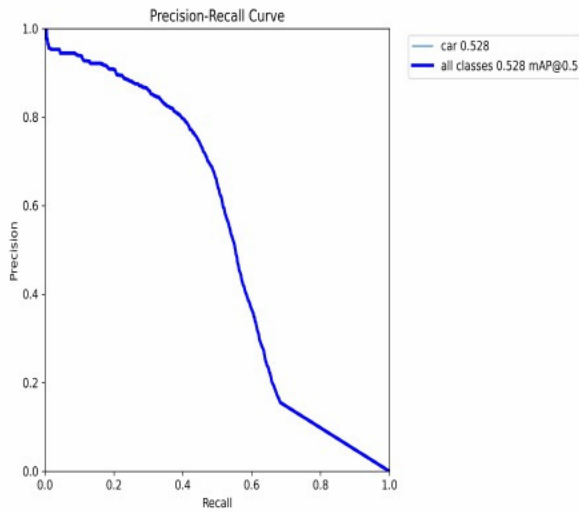


Figure 7: Precision-Recall curve of the student model after teacher-student adaptation.

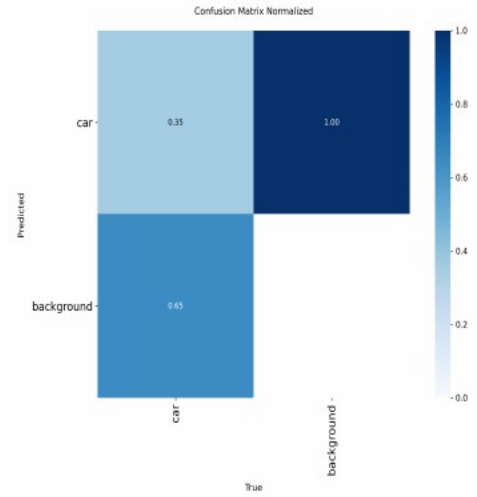


Figure 8: Normalized confusion matrix of the adapted student model on real-world validation data.

5.4 Qualitative Evaluation and Failure Analysis

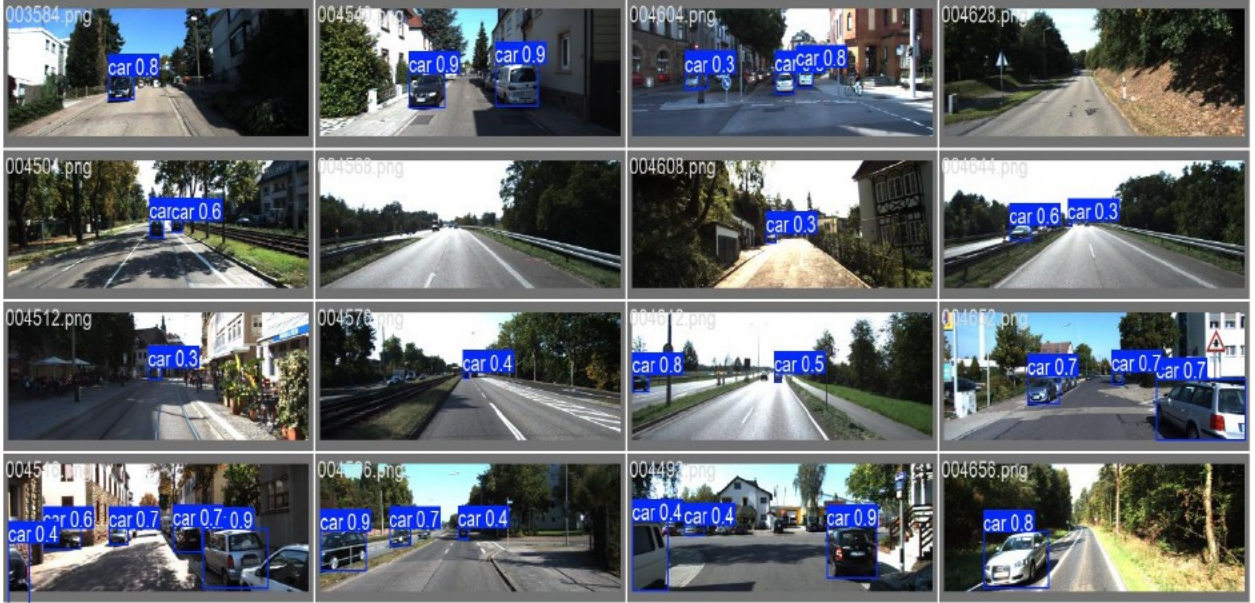


Figure 9: Teacher model predictions on real target-domain images used for pseudo-label generation.



Figure 10: Qualitative detection results of the adapted student model on real driving scenes.

Qualitative results on real-world driving images, such as those shown in Figure 9 and Figure 10, indicate that the self-trained student model is indeed able to detect cars across a wide range of real-world scenarios with varied lighting conditions, occlusions, and complex backgrounds. This reinforces the intuition that even a relatively simple self-training pipeline without adversarial domain alignment or complex augmentation strategies is often able to make good use of real, unlabeled data to attain meaningful performance gains.

However, as pointed out by previous works, self-training has its own drawbacks. The error in pseudo-labeling propagates: incorrect pseudo-labels (false positives or imprecise boxes) may reinforce the model’s biases. Recent works try to mitigate this via stronger regularization, better pseudo-label filtering, or domain-specific adaptation strategies. For example, the paper Simplifying Source Free Domain Adaptation for Object Detection investigates how

subtle design choices, such as batch-normalization adaptation and augmentation strategies, affect the performance of self-training-based domain adaptation. [14]

6 Conclusion

This work explored three complementary domain adaptation strategies, lightweight GAN-based image translation, feature-level alignment through a DANN-style architecture, and a teacher–student self-training framework to address the synthetic-to-real performance gap in car detection. Although each method targets a different source of domain shift, our experiments showed that the teacher–student strategy provided the most meaningful improvements under limited computational resources.

The lightweight GAN model demonstrated that appearance translation can increase precision but often degrades structural details, particularly for small objects such as cars. This emphasizes a common challenge in image-level adaptation, visually realistic outputs do not always translate into more informative training data. The attempted DANN integration revealed practical difficulties in modifying the internal YOLO architecture to support adversarial alignment, highlighting the engineering complexity of feature-level domain adaptation in modern detection frameworks. In contrast, the self-training pipeline that combines synthetic labeled data with high-confidence pseudo-labels from real images produced a clear improvement in real-domain performance. The adapted student model achieved higher mAP, precision, and qualitative robustness compared to the synthetic-only baseline. This shows that even a simplified teacher-student setup can effectively use unlabeled real-world data.

Nonetheless, limitations remain. Recall on real images is still constrained, and pseudo-labeling errors may propagate during training. Future work could integrate stronger augmentations, more selective pseudo-label filtering, or hybrid strategies that combine self-training with lightweight adversarial alignment or normalization adaptation. These extensions may further reduce domain shift while maintaining computational efficiency.

References

- [1] Konstantinos Bousmalis, Nathan Silberman, David Dohan, Dumitru Erhan, and Dilip Krishnan. Unsupervised pixel-level domain adaptation with generative adversarial networks. In *CVPR*, 2017.
- [2] Ian Goodfellow. NIPS 2016 tutorial: Generative adversarial networks. *arXiv preprint arXiv:1701.00160*, 2016.
- [3] Alireza Makhzani, Jonathon Shlens, Navdeep Jaitly, Ian Goodfellow, and Brendan Frey. Adversarial autoencoders. In *ICLR*, 2016.
- [4] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *ICCV*, 2017.
- [5] Matthew Johnson-Roberson, Charles Barto, Rounak Mehta, Sharath Nittur Sridhar, Karl Rosaen, and Ram Vasudevan. Driving in the matrix: Can virtual worlds replace human-generated annotations for real world tasks? *arXiv preprint arXiv:1610.01983*, 2016.
- [6] Andreas Geiger, Philip Lenz, and Raquel Urtasun. Are we ready for autonomous driving? The KITTI vision benchmark suite. In *CVPR*, 2012.
- [7] Diederik P. Kingma and Max Welling. Auto-Encoding Variational Bayes. *arXiv preprint arXiv:1312.6114*, 2013.
- [8] Calvin Luo. Understanding Diffusion Models: A Unified Perspective. *arXiv preprint arXiv:2208.11970*, 2022.
- [9] Xingyi Yang, Daquan Zhou, Jiashi Feng, and Xinchao Wang. Diffusion Probabilistic Model Made Slim. *arXiv preprint arXiv:2211.17106*, 2022.
- [10] Xingchao Peng, Ben Usman, Kuniaki Saito, Neela Kaushik, Judy Hoffman, and Kate Saenko. Syn2Real: A New Benchmark for Synthetic-to-Real Visual Domain Adaptation. *arXiv preprint arXiv:1806.09755*, 2018.
- [11] Yufei Liu, et al. Generative Modeling for Small-Data Object Detection. *arXiv preprint arXiv:1910.07169*, 2019.
- [12] Yen-Cheng Liu, Chih-Yao Ma, Zijian He, Chia-Wen Kuo, Kan Chen, Peizhao Zhang, Bichen Wu, Zsolt Kira, and Peter Vajda. Unbiased Teacher for Semi-Supervised Object Detection. *arXiv preprint arXiv:2102.09480*, 2021.
- [13] Yu-Jhe Li, Xiaoliang Dai, Chih-Yao Ma, Yen-Cheng Liu, Kan Chen, Bichen Wu, Zijian He, Kris Kitani, and Peter Vajda. Cross-Domain Adaptive Teacher for Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7581–7590, 2022.
- [14] Yan Hao, Florent Forest, and Olga Fink. Simplifying Source-Free Domain Adaptation for Object Detection: Effective Self-Training Strategies and Performance Insights. In *European Conference on Computer Vision*, pages 196–213, 2024.
- [15] Mehran Khodabandeh and Arash Vahdat and Mani Ranjbar and William G. Macready. A Robust Learning Approach to Domain Adaptive Object Detection *arXiv preprint arXiv:1904.02361*, 2019.
- [16] Shizhao Zhang, Hongya Tuo, Jian Hu, Zhongliang Jing. Domain Adaptive YOLO for One-Stage Cross-Domain Detection *arXiv preprint arXiv:2106.13939*, 2021.
- [17] Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi. You Only Look Once: Unified, Real-Time Object Detection *arXiv preprint arXiv:1506.02640*, 2015.
- [18] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, Victor Lempitsky. Domain-Adversarial Training of Neural Networks *arXiv preprint arXiv:1505.07818*, 2015.
- [19] Zhang, Ying and Sun, Hong-Mei and Yang, Rui and Jia, Rui-Sheng. UDANet: An unsupervised domain adaptive vehicle density estimation network based on joint adversarial learning. In *Pergamon Press, Inc.*, 2024.
- [20] Yuhua Chen, Wen Li, Christos Sakaridis, Dengxin Dai, Luc Van Gool. Domain Adaptive Faster R-CNN for Object Detection in the Wild *arXiv preprint arXiv:1803.03243*, 2018.